Kunal Chevuri

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High school senior researching federated learning, vision-language models, LLM evaluation and network analysis at UT Rio Grande Valley, George Mason, UNT and ThinkNeuro. Six published or accepted IEEE papers; four under review.

EDUCATION

Prosper High School **Prosper, TX Aug 2023 – May 2027 (expected)**
GPA 5.363/6.0 weighted (4.0 unweighted) | Rank 15/637 | SAT 1540 (Math 800, Reading & Writing 740), Dec 2025

- **AP scores:** 5 in Precalculus, Computer Science A and Human Geography; 4 in Calculus BC, Statistics, Physics 1, Microeconomics, U.S. History, Computer Science Principles and World History

RESEARCH PUBLICATIONS

Published / Accepted for Publication

[1] **K. Chevuri**, "Visual Degradation Breaks a Fine-Tuned Vision-Language Model but Not a Frozen Self-Supervised Encoder: A Robustness Study for Roadway Flood Monitoring," in *Proc. IEEE MIT Undergraduate Research Technology Conf. (URTC)*, 2026. Sole author. Code

[2] **K. Chevuri**, A. Peredo, and S. Chuprov, "Diagnosing the Synthetic-to-Real Gap in Small VLMs for Deployable Flood Monitoring," accepted to *IEEE Int. Conf. Cognitive Machine Intelligence (CogMI)*, 2026. Code | Dataset (Zenodo)

[3] **K. Chevuri**, A. Poon, R. Mahabir, et al., "An Open Deal-Level Dataset of African Agrifood Investment, 2015–2025, and the Reproducible Pipeline That Builds It," accepted to the *High School Symposium, IEEE Int. Conf. Big Data*, 2026. Joint first author. Code & Data

[4] **K. Chevuri** and S. Chuprov, "Masking or Noise? Null-Calibrated Auditing of Per-Dimension Failures Behind Composite Wellness Scores in Federated Learning," accepted to the *ARRL-Health Workshop, IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, 2026. Code

[5] **K. Chevuri**, L. Ileto, and M. Aledhari, "Auditing Data Leakage and Reporting Gaps in EEG Connectivity Deep Learning for Depression Detection," accepted to the *7th Int. Workshop on Machine Learning for EEG Signal Processing (MLESP), IEEE BIBM*, 2026.

[6] **K. Chevuri**, A. Sheikh, and M. Aledhari, "Measuring Evasion in Institutional Question-Answering: A Training-Free LLM-as-Judge Framework with Cross-Domain Validation," accepted to the *3rd IEEE Int. Workshop on Large Language Models for Finance, IEEE Int. Conf. Big Data*, 2026. Code

Submitted / Under Review

[7] **K. Chevuri** and S. Chuprov, "Robustness or Weighting? Re-Evaluating Federated Aggregation on Heterogeneous Health Data," submitted to the *2nd Workshop on AI in Big Data Analytics for Healthcare, IEEE Int. Conf. Big Data*, 2026. Code

[8] **K. Chevuri**, A. Poon, R. Mahabir, et al., "Do Development Finance Institutions Crowd In Private Capital? Network Evidence from African Agrifood Investment, 2015–2025," submitted to *World Development*, 2026. Joint first author.

[9] **K. Chevuri** and M. Aledhari, "Predicting U.S. Public-Company Failure from Free Regulatory Filings: A Reproducible Pipeline, an Operating-Point Analysis, and an Audit of Its Own Failure Modes," submitted to *Expert Systems with Applications*, 2026. First author.

[10] **K. Chevuri** and M. Aledhari, "Decomposing the Post-2020 Rise in the Cancer Multiple-Cause to Underlying-Cause Ratio: COVID-Involved and Non-COVID Components, United States 2003–2024," submitted to *Demographic Research*, 2026. First author.

Conference Presentations & Posters

[11] Paper [7] presented at *University of North Texas Research Day* (Oct. 1, 2026) and the *National Research Conference*, University of Pennsylvania (Oct. 2–3, 2026).

[12] Paper [10] presented at the *Texas Junior Academy of Science* (Oct. 8, 2026).

[13] **K. Chevuri**, L. Ileto, et al. (11 authors), "Deep Learning and EEG Functional Connectivity for Major Depressive Disorder: A Bibliometric Analysis of Research Trends and Validation Practices," e-poster, *ThinkNeuro Cognitive and Computational Neuroscience Program*, 2026. First author.

Under Preparation

[14] Startup survival among newly public technology companies, testing whether patent-backed technology and AI at initial public offering predict survival beyond financial measures alone, for *Technological Forecasting and Social Change*. First author.

RESEARCH EXPERIENCE

ThinkNeuro, Cognitive and Computational Neuroscience Program **May 2026 – Present**
Project Manager (Aug 2026 – Present); Associate Research Intern (May – Aug 2026)

- Only high school student appointed project manager in a cohort of 1,000+ interns; manage 5 research groups.
- Led 11 researchers through a review of 100 published studies on EEG and deep learning for major depressive disorder; first author of the e-poster [13] and the companion data-leakage audit [5].

George Mason Univ. & Alliance for a Green Revolution in Africa (AGRA) **Apr 2026 – Present**
Research Intern, Aspiring Scientists Summer Internship Program (≈9% acceptance)

- Built and audited the open deal-level database behind the first network study of development finance in African agrifood systems: 1,060 investor–deal records across 389 transactions, 420 investors, 313 companies and 30 countries from 121 sources; personally verified 87.2% against a primary source [3].
- Wrote the cleaning, company name-matching and audit software that rebuilds it identically on every run.

- Found that public development banks hold 6.04× the brokerage centrality of other investors ($p < 0.0001$) yet partner with private investors at only 1.51× chance vs. 4.92× for private pairs [8]. Shared findings with 20+ African government officials; the work will be credited in AGRA's 2027 Africa Food Systems Report.

University of North Texas **Apr 2026 – Present**
Student Researcher; advised by Dr. Mohammed Aledhari

- Built and validated an LLM-as-judge scoring evasion in 3,350 SEC earnings-call exchanges ($r = 0.78$–0.80 against 2 human raters plus 2 external corpora); found that evasion scores did not predict 3-day abnormal stock returns after the call, reported that result, and released the pipeline at $0.0038 per exchange [6].
- Built a company-failure prediction system on a 57,000 company-year panel (69 measures from free government filings), auditing labels for contamination and tracking accuracy through COVID [9]; extended it into a startup feasibility model reporting failure risk and growth as separate scores [14].

Robust Intelligent Systems Lab, Univ. of Texas Rio Grande Valley **Mar 2026 – Present**
Student Researcher; advised by Dr. Sergei Chuprov

- Ran 624 experiments across 7 federated-learning methods on 382,709 CDC records and showed a simple averaging baseline recovers 79–98% of the advantage each complex method claims [7]; wrote a 181-check script that re-derives every reported number from the raw results.
- Created the Masking Gap, a chance-calibrated measure of how far a composite wellness score hides a person's worst individual health problem, on 8,353 NHANES records [4].
- Generated 1,005 synthetic flood images from real traffic-camera frames, released under a permanent DOI; a frozen I-JEPA encoder reached 80.8% macro-recall while fine-tuned VLMs transferred poorly [1], [2].

HONORS, AWARDS & COMPETITIONS

- **SkillsUSA Career Pathways Showcase:** National Top 10, 1st Texas, 1st District (2026); 5th Texas, 1st District (2025)
- **National Merit Commended Scholar** (Aug 2026) | **AP Scholar with Distinction** (Jul 2025) | **National Honor Society** (Fall 2025)
- **DECA Entrepreneurship Series:** Texas State Qualifier (Jan 2026) | **Presidential AI Challenge:** national competitor (Jan 2026) | **Gold President's Volunteer Service Award:** 220+ hours (Nov 2024)
- **Soccer:** Solar SC ECNL Academy – ECNL Texas Conference Champions, National Top 16 (2023–24), Dallas Cup Finalists, SuperCopa+ Champions (2024), ECNL National Top 8 (2024–25) | Prosper HS Varsity – Academic All-District (2023–24, 2024–25, 2025–26), Honorable Mention All-District (2025–26)

LEADERSHIP, ENTREPRENEURSHIP & SERVICE

Hope4Spandana, Texas Chapter **Mar 2024 – Present**
Co-Founder and President

- Grew the chapter from 6 to 100+ members and raised $13,000+ across 20+ fundraisers, partnering with 10+ businesses and nonprofits (North America Telugu Society, EkamUSA, Patel Brothers, Desi District, Swadeshi).
- Directed proceeds to a shelter housing 100 people, 20+ storefronts owned by physically disabled entrepreneurs, and a school providing free education to 75 children with autism, in India.

SkillsUSA, Prosper High School and Prosper ISD **Aug 2024 – Present**
Chapter Vice President (May 2026 – Present); PR Committee Head (Sep 2025 – Present); District Reporter (May 2025 – Present)

- Grew the chapter by 70+ members, led fundraising of roughly $5,000, ran district-wide communications and organized 10+ career-building workshops.
- Led 3-person teams to National Top 10 with the Energy Blackout Risk Monitoring Platform, which forecasts grid blackout risk from public energy data (2026), and to 5th in Texas with a 3D radar detection system (2025).

Prosper HS Artificial Intelligence and Data Science Club **Sep 2025 – Present**
Co-Founder and Vice President

- Co-founded the school's first data science organization with an introductory curriculum assuming no prior programming; grew it to 40+ members and mentored four 3-person teams through DataJam and the Congressional App Challenge.

Rayfield Systems, *Machine Learning Intern and Team Lead* (Jun – Jul 2025): placed through the Harvard Undergraduate Venture Tech Summer Program (≈15% acceptance); led a 6-person team building a Python anomaly-detection, forecasting and clustering tool for energy data; pitched it to the CEO and the company adopted it.

Independent Soccer Coaching Business, *Founder and Head Coach* (Jun – Aug 2025; Jun – Aug 2026): ran 10+ revenue-generating camps of 10+ players each and coached 5+ players one-on-one, priced below local club rates.

Prosper HS Solar Car Engineering Team, *CAD Team Lead* (Aug 2023 – Apr 2024): led the CAD subteam, producing the Fusion 360 models used in the vehicle design.

SKILLS & CERTIFICATIONS

Languages & Tools: Python (primary research language), Java (AP CSA), R (certified), Unix shell, Git, LaTeX, Jupyter, RStudio, Linux/HPC (NVIDIA A100), Fusion 360, Excel. Published code: six public repositories at github.com/kunalchevuri, each shipping a verification script (181- and 278-check) that re-derives every number in the paper from the raw results

ML & Deep Learning: PyTorch, Hugging Face Transformers, PEFT (LoRA/QLoRA), 4-bit quantization, scikit-learn, XGBoost, SHAP, Flower, Pandas, NumPy, NetworkX, Matplotlib; SmolVLM2, Qwen2-VL, I-JEPA, SigLIP/ViT, linear probing; FedAvg, FedProx, SCAFFOLD, Ditto, q-FedAvg, Krum, trimmed mean, non-IID partitioning; LLM-as-judge evaluation

Statistics & Data: Mann–Whitney, Wilcoxon, McNemar, Cohen's κ, bootstrap CIs, permutation and configuration-model nulls, Louvain, AUROC, power analysis; SEC EDGAR/XBRL, CDC BRFSS, NHANES, CDC WONDER. **Certification:** R and Data Analytics, ThinkNeuro (Apr 2026). **Memberships:** NHS, SkillsUSA, DECA